Sorting Programs in C

Quick Sort • Merge Sort • Heap Sort • Insertion Sort • Bubble Sort

Bubble Sort

Har pair ko compare karke bade ko aage bhejte hain. Sabse simple sort hai.

```c
// ================================================
// BUBBLE SORT - C Program
// ================================================
#include <stdio.h>

int main()
{
    int a[20], n, i, j, temp;

    printf("Enter number of elements: ");
    scanf("%d", &n);

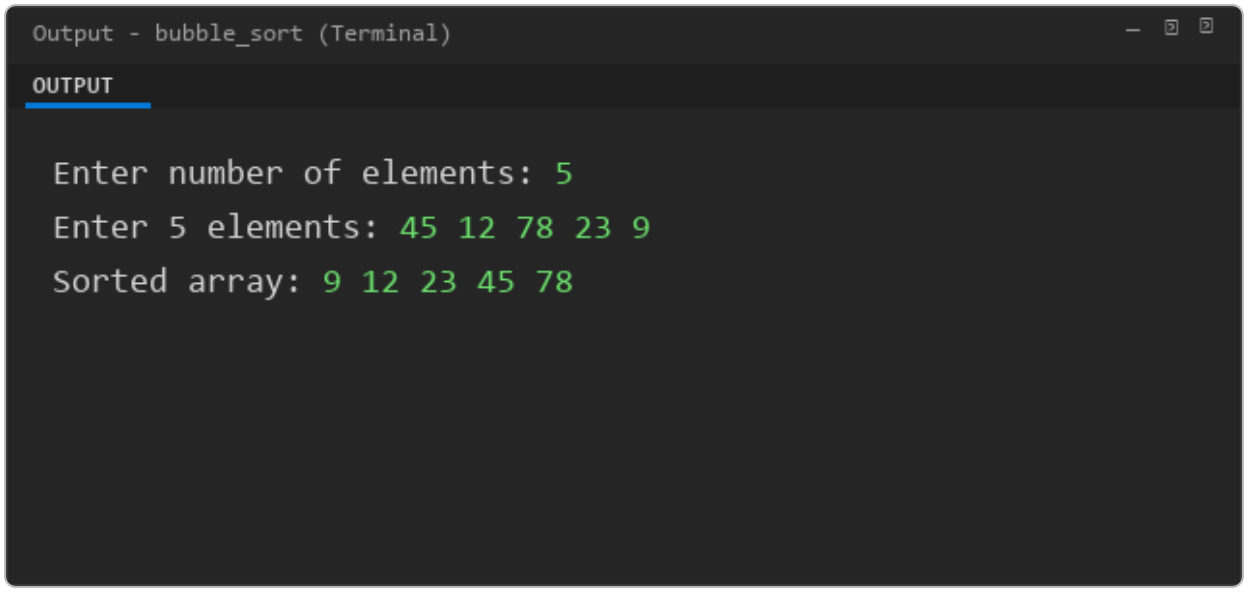
    printf("Enter %d elements: ", n);
    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);

    // Bubble sort logic
    for (i = 0; i < n - 1; i++)
    {
        for (j = 0; j < n - i - 1; j++)
        {
            if (a[j] > a[j + 1])
            {
                temp = a[j];
                a[j] = a[j + 1];
                a[j + 1] = temp;
            }
        }
    }

    printf("Sorted array: ");
    for (i = 0; i < n; i++)
        printf("%d ", a[i]);
```

```
    return 0;
}
```

Output (VS Code Terminal Screenshot):



Insertion Sort

Cards jaise element ko uthakar sahi jagah par daal dete hain.

```c
// ================================================
// INSERTION SORT - C Program
// ================================================
#include <stdio.h>

int main()
{
    int a[20], n, i, j, key;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    printf("Enter %d elements: ", n);
    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);

    // Insertion sort logic
    for (i = 1; i < n; i++)
    {
        key = a[i];
```

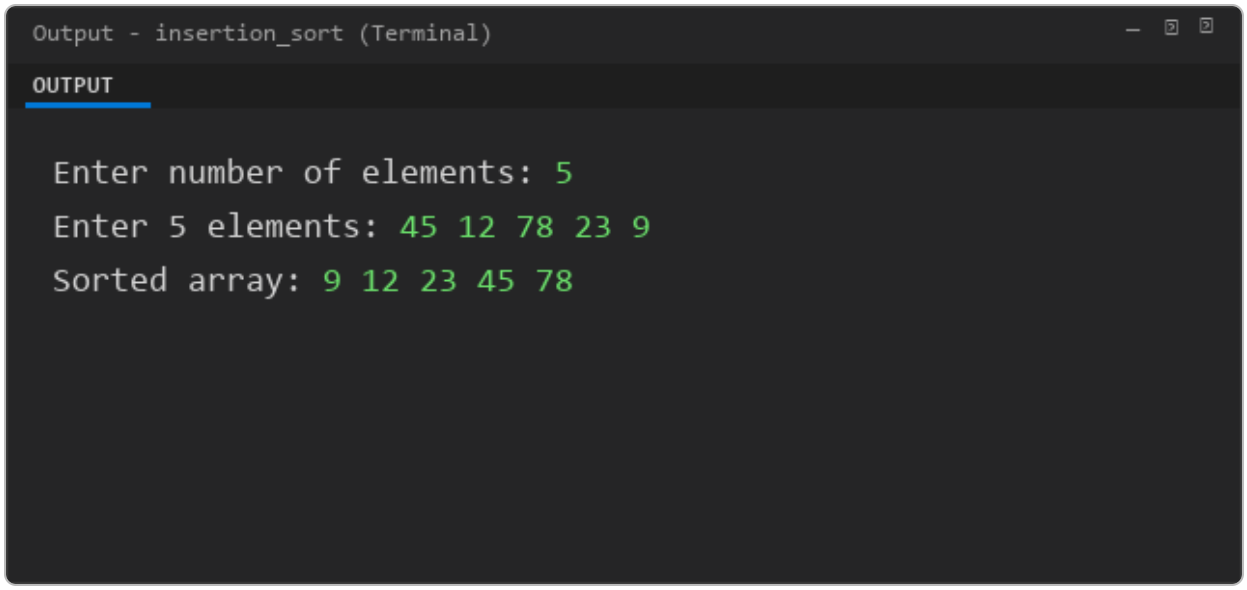
```
        j = i - 1;

        while (j >= 0 && a[j] > key)
        {
            a[j + 1] = a[j];
            j = j - 1;
        }
        a[j + 1] = key;
    }

    printf("Sorted array: ");
    for (i = 0; i < n; i++)
        printf("%d ", a[i]);

    return 0;
}
```

Output (VS Code Terminal Screenshot):



Quick Sort

Pivot chunke uske chhote left, bade right me daalte hain. Sabse fast.

```
// ================================================
// QUICK SORT - C Program
// ================================================
#include <stdio.h>

// Partition function: pivot ke left me chhote, right me bade
```

```c
int partition(int a[], int low, int high)
{
    int pivot = a[high];
    int i = low - 1;
    int j, temp;

    for (j = low; j < high; j++)
    {
        if (a[j] < pivot)
        {
            i++;
            temp = a[i];
            a[i] = a[j];
            a[j] = temp;
        }
    }
    temp = a[i + 1];
    a[i + 1] = a[high];
    a[high] = temp;

    return i + 1;
}

// Quick sort function
void quickSort(int a[], int low, int high)
{
    int p;

    if (low < high)
    {
        p = partition(a, low, high);
        quickSort(a, low, p - 1);
        quickSort(a, p + 1, high);
    }
}

int main()
{
    int a[20], n, i;

    printf("Enter number of elements: ");
    scanf("%d", &n);

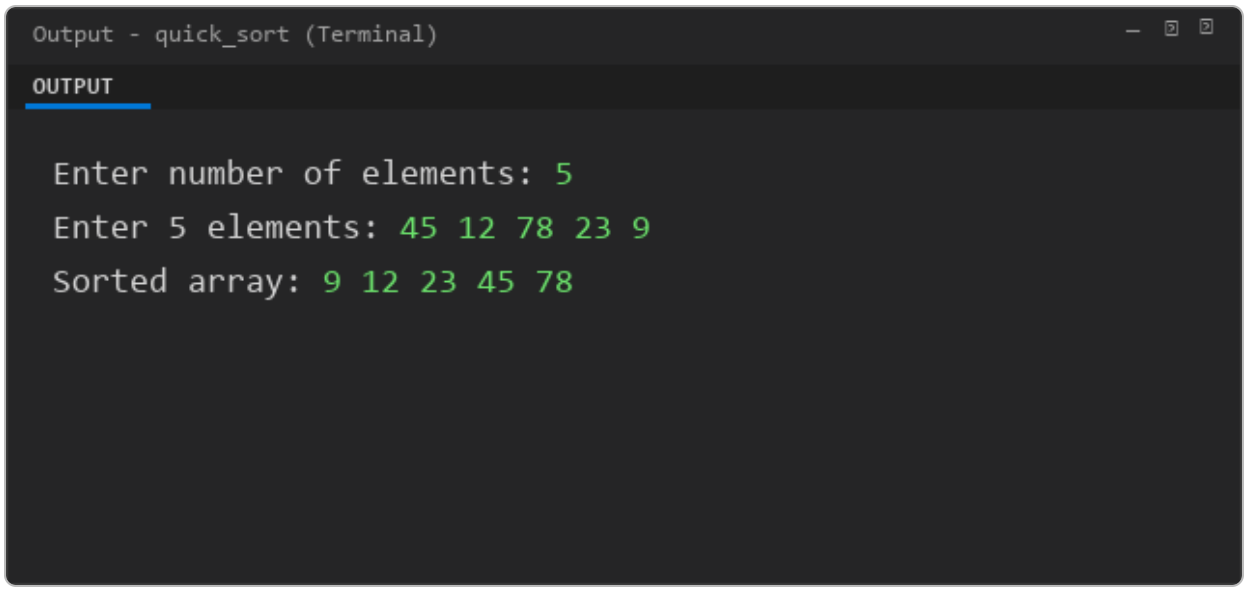
    printf("Enter %d elements: ", n);
    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);

    quickSort(a, 0, n - 1);
```

```c
    printf("Sorted array: ");
    for (i = 0; i < n; i++)
        printf("%d ", a[i]);

    return 0;
}
```

Output (VS Code Terminal Screenshot):



Merge Sort

Array ko 2 hisso me todte hain, sort karke wapas jodte hain.

```c
// ================================================
// MERGE SORT - C Program
// ================================================
#include <stdio.h>

// Do sorted parts ko merge karo
void merge(int a[], int left, int mid, int right)
{
    int b[20];
    int i = left, j = mid + 1, k = left;

    while (i <= mid && j <= right)
    {
        if (a[i] <= a[j])
            b[k++] = a[i++];
```

```
        else
            b[k++] = a[j++];
    }

    while (i <= mid)
        b[k++] = a[i++];

    while (j <= right)
        b[k++] = a[j++];

    for (i = left; i <= right; i++)
        a[i] = b[i];
}

// Merge sort function
void mergeSort(int a[], int left, int right)
{
    int mid;

    if (left < right)
    {
        mid = (left + right) / 2;
        mergeSort(a, left, mid);
        mergeSort(a, mid + 1, right);
        merge(a, left, mid, right);
    }
}

int main()
{
    int a[20], n, i;

    printf("Enter number of elements: ");
    scanf("%d", &n);

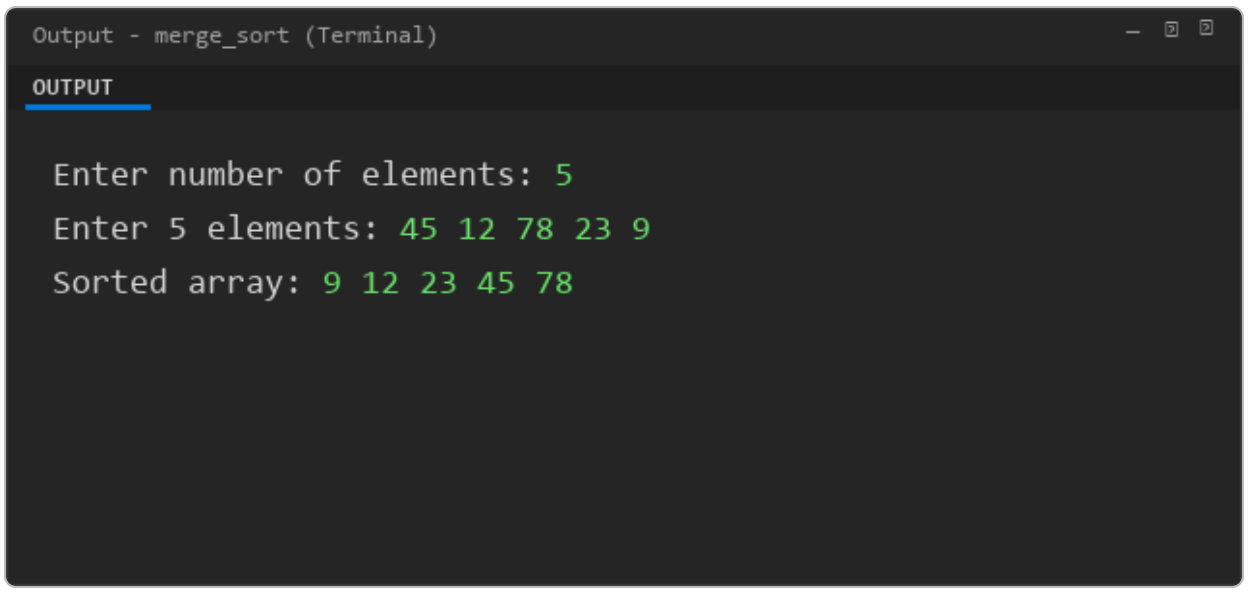
    printf("Enter %d elements: ", n);
    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);

    mergeSort(a, 0, n - 1);

    printf("Sorted array: ");
    for (i = 0; i < n; i++)
        printf("%d ", a[i]);

    return 0;
}
```

Output (VS Code Terminal Screenshot):



Heap Sort

Max-heap banakar sabse bada element ek-ek karke nikalte hain.

```c
// ================================================
// HEAP SORT - C Program
// ================================================
#include <stdio.h>

// Max-heap banane ka function
void heapify(int a[], int n, int i)
{
    int largest = i;
    int left = 2 * i + 1;
    int right = 2 * i + 2;
    int temp;

    if (left < n && a[left] > a[largest])
        largest = left;

    if (right < n && a[right] > a[largest])
        largest = right;

    if (largest != i)
    {
        temp = a[i];
        a[i] = a[largest];
        a[largest] = temp;
```

```
        heapify(a, n, largest);
    }
}

// Heap sort function
void heapSort(int a[], int n)
{
    int i, temp;

    // Step 1: max heap banao
    for (i = n / 2 - 1; i >= 0; i--)
        heapify(a, n, i);

    // Step 2: ek-ek karke root nikalo
    for (i = n - 1; i > 0; i--)
    {
        temp = a[0];
        a[0] = a[i];
        a[i] = temp;

        heapify(a, i, 0);
    }
}

int main()
{
    int a[20], n, i;

    printf("Enter number of elements: ");
    scanf("%d", &n);

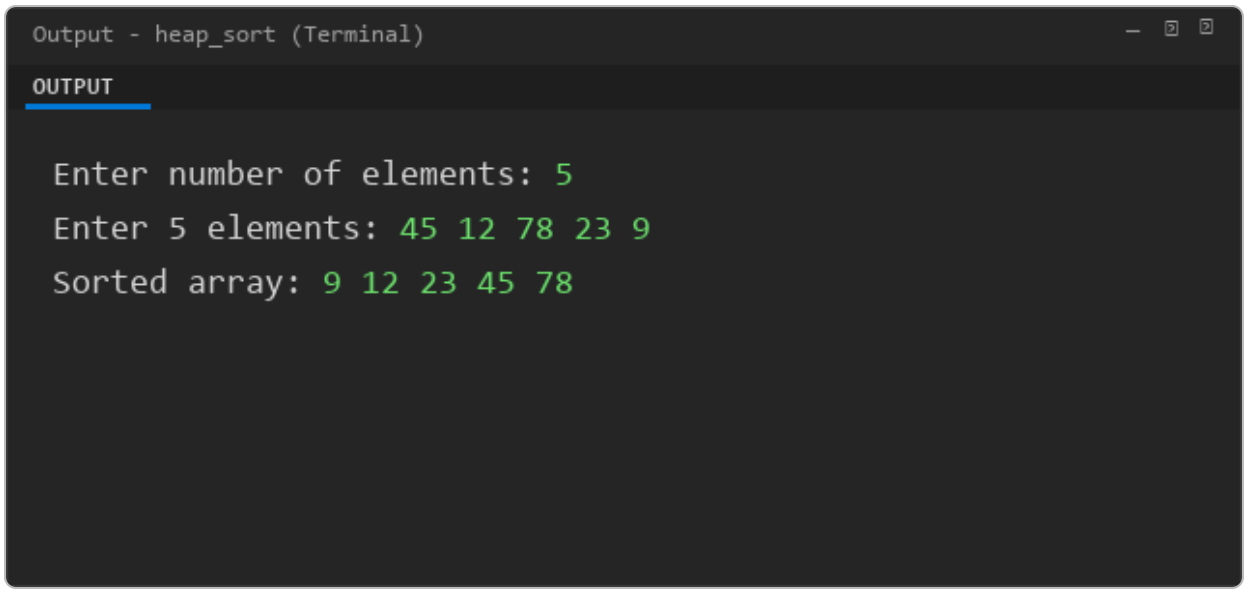
    printf("Enter %d elements: ", n);
    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);

    heapSort(a, n);

    printf("Sorted array: ");
    for (i = 0; i < n; i++)
        printf("%d ", a[i]);

    return 0;
}
```

Output (VS Code Terminal Screenshot):


Output - heap_sort (Terminal)
OUTPUT
Enter number of elements: 5
Enter 5 elements: 45 12 78 23 9
Sorted array: 9 12 23 45 78